HANYEOL RYU

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RESEARCH INTERESTS

- Low-Power Wireless Communications (i.e. Backscatter Communications)
- Channel Modeling & Estimation
- AI/ML for Intelligent Wireless Communications
- Self-Sustainable mmWave Communication and Sensing Systems
- Wireless Security & Privacy

EDUCATION

Pusan National University

- *Master of Science (M.S.) in Electrical and Electronics Engineering* 2026 – Present
- *Bachelor of Science (B.S.) in Electronics Engineering* 2021 – 2026

RESEARCH EXPERIENCE

Pusan National University

- *Research Assistant (PI: Dr. Sangkil Kim)* 2024 – Present

PROFESSIONAL ACTIVITIES

Technical Program Committee (TPC) Reviewer

- IEEE ISPACS 2026

Journal Reviewer

- IEEE Internet of Things Journal (IoT-J) 2026

PUBLICATIONS

1. **Hanyeol Ryu**, Sangkil Kim, "Ultralow-Power Monostatic Backscatter Platform With Phase-Aware Channel Estimation and System-Level Validation," *IEEE Internet of Things Journal*, 2026. (DOI: 10.1109/JIOT.2026.3696951)
2. Nohgyeom Ha, **Hanyeol Ryu**, Hyunmin Jeong, Hoyong Kim, Gyoungdeuk Kim, Apostolos Georgiadis, Manos M. Tentzeris, Sangkil Kim, "Integrated Sensing and Backscatter Communication: Toward Self-Sustainable Autonomous Internet of Things Networks," *IEEE Microwave Magazine,* 2026. (DOI: 10.1109/MMM.2026.3717878)
3. **Hanyeol Ryu**, Nohgyeom Ha, Sangkil Kim, "Transformer-Hypernetwork-Controlled Deep-Unfolded Phase-Aware Channel Estimation Refinement for Phase-Drift-Robust Backscatter Links," *arXiv preprint,* 2026. (DOI: 10.48850/arXiv.2606.31400).

4. **<u>Hanyeol Ryu</u>**, Sangkil Kim, "Deep Learning-Aided Phase-Aware Channel Estimation for Error-Floor Suppression over Phase-Drifting Backscatter Links," *IEEE International Symposium on Intelligent Signal Processing and Communication Systems (ISPACS)*, Accepted, 2026.
5. **<u>Hanyeol Ryu</u>**, Sangkil Kim, "High-Isolation Yagi-Uda Antenna Array using Split-Ring Resonators for Practical Backscatter System," *IEEE International Symposium on Antennas and Propagation and USNC-URSI Radio Science Meeting (AP-S/URSI)*, 2025. (DOI: 10.1109/AP-S/CNC-USNC-URSI55537.2025.11266818)
6. Jeonghyeon Choi[1], **<u>Hanyeol Ryu</u>[1]**, Sangkil Kim, "Improved Backscattering Communication Range Using Reflection Amplifiers," *The Journal of Korean Institute of Electromagnetic Engineering and Science,* 2025. (DOI: 10.5515/KJKIEES.2025.36.3.266)

PATENT

- Korean Patent Application, allowed April 2026. "Apparatus and Method for Phase-Aware Channel Estimation in Monostatic Backscatter Communication Systems." <u>Hanyeol Ryu</u>, Sangkil Kim.

CONFERENCE PRESENTATIONS

Title: *Deep Learning-Aided Phase-Aware Channel Estimation for Error-Floor Suppression over Phase-Drifting Backscatter Links*

- IEEE ISPACS — Oct 2026

Title: *Ultralow-Power Backscatter Platform With Phase-Aware Channel Estimation*

- The Korean Institute of Electromagnetic Engineering and Science (KIEES) — Feb 2026

Title: *High-Isolation Yagi-Uda Antenna Array using Split-Ring Resonators for Practical Backscatter System*

- IEEE AP-S/URSI — Jul 2025

Title: *Improved Backscattering Communication Range Using Reflection Amplifiers*

- KIEES — Feb 2025

HONORS & AWARDS

- **Excellence Award (2nd Place)**, Hackathon, Pusan National University — Aug 2023
- **Excellence Award (2nd Place)**, Capstone Design Project, Pusan National University — Nov 2025

SERVICE & EXTRACURRICULAR ACTIVITIES

<u>University</u>

Pusan National University

- Treasurer, Department Student Council — Dec 2021 – Dec 2022
- Student Ambassador, Pusan National University — Apr 2021 – Feb 2023
- Volunteer, Overseas Volunteering Program (3D Printer & Arduino) — Thailand — Dec 2022 – Feb 2023

Military Service

Republic of Korea Marine Corps (ROKMC)

- Squad Leader, ROKMC Force Reconnaissance (Force Recon) 2017 – 2018
- 2^{nd} Place, ROKMC Force Recon Qualification Course 2017

SKILLS & TECHNIQUES

- **Programming:** MATLAB, Python, C++
- **AI/ML:** PyTorch, TensorFlow/Keras
- **RF/EM Simulation:** Ansys HFSS, Advanced Design System (ADS)
- **Software-Defined Radio (SDR):** GNU Radio
- **Hardware Design:** OrCAD PCB Editor, OrCAD Capture
- **3D Modeling:** Rhino 7